Karaayvas 2018 TNBC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample: per-neighbourhood same-sample fraction 0.82→0.27 (chance 0.20); iLISI 1.5→4.1 of 5 samples. One connected component, 0 detached islands. Five samples intermix across a single continuous cloud — GO.

Gate 2 — module geography → NO-GO

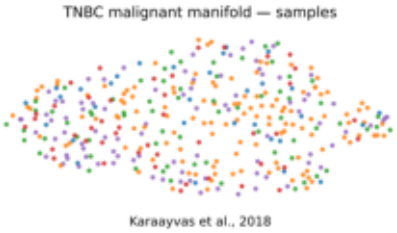
Module geography is ABSENT in the raw per-cell scores (Gast~pt ρ=−0.08, pSom +0.02, Crest −0.03) and appears only after graph-smoothing. A permutation null (360 cells) gives 95% CI ±0.9; every smoothed gradient is non-significant (p≥0.26). Tell: smoothing FLIPS the sign of the one real raw signal (Crest↔Gast −0.43→+0.44, Crest↔pSom −0.55→+0.10) — fabricated, not latent. Pattern absent.

Gate 3 — cross-repression → NA

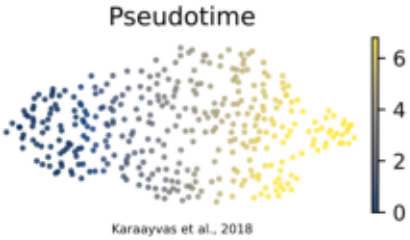
Non-branching topology (curve: 2 tips, 0 forks across N=15/20/30; PPT tree forks unstable 1–4 = spurious in a diffuse cloud). No bifurcation to test — NA.

STEP FIGURES

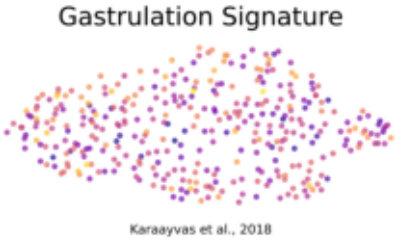
Manifold — samples



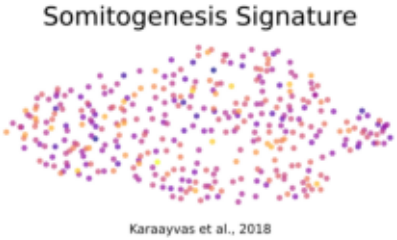
Pseudotime



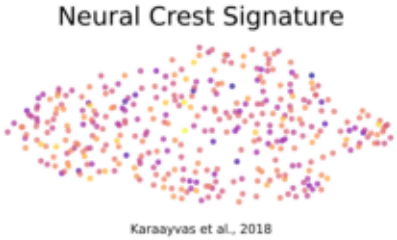
Gastrulation



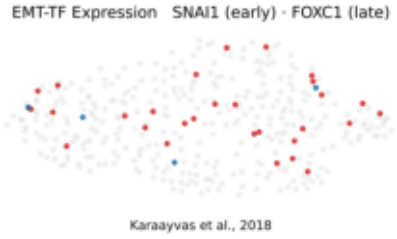
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (TNBC) (Karaayvas et al., 2018), SmartSeq2 — 365 malignant cells (6 samples reported) → 360 analysed (5 samples ≥10 cells). Sparse plate-based data. A trajectory is fit, but the module pattern is absent (Gate 2 NO-GO).

Verdict.
Pattern does NOT reproduce. Gate 1 holds (360 cells intermix on one continuous manifold), but the embryonic-module geography is absent at single-cell resolution: raw pseudotime gradients are ~0 and the smoothed structure is a graph-smoothing artifact (null 95% CI ±0.9, all p≥0.26; smoothing flips the raw Crest signs). Gate 2 NO-GO, Gate 3 NA. EMT-TFs SNAI1 (early/root) → FOXC1 (late/terminus).

Parameter choices

Threshold	≥10 cells/sample (SmartSeq2 scale; floor 100 keeps only 1/6); cliff 31→5 → 5/6, 360 cells	**Embedding**	Harmony θ=10; n_pcs=20, n_neighbors=30, spread=1.3; 51 HVG
Trajectory	scFates curve (non-branching), N=20, λ=0.3, μ=100; fold-free (2.97)	**Root**	Gast/pSom-high, Crest-low tip; not proliferation-rooted (pt–cycling ρ=+0.18)
Geometry	Gate 2 NO-GO — module pattern absent at single-cell resolution	**EMT-TFs**	SNAI1 (early/root) → FOXC1 (late/terminus); 10-TF panel, VIM excluded